EMBEDDED / ELECTRONICS / ROBOTICS

CUSTOM AUTONOMOUS DRONE

01

EXISTING BUILD / EVIDENCE TO CURATE

THE ENGINEERING QUESTION

How do separately designed power, airframe, control and software subsystems become one verifiable flying system?

PROBLEM DEFINITION AND RESEARCH CONTRIBUTION

A real end to end autonomous aircraft combining mechanical design, PCB engineering, embedded flight control, sensing, telemetry and an operator application.

- Document the original build from real artefacts rather than retrospective marketing copy.
- Separate what was actually tested on hardware from what existed in simulation or concept form.
- Use this as the anchor project for embedded, electronics, robotics and systems roles.

ARCHITECTURE AND IMPLEMENTATION SPECIFICATION

1 STAGE
Airframe & power

→

2 STAGE
Flight firmware

→

3 STAGE
State estimation & control

→

4 STAGE
Vision / telemetry / app

- Power management design, board revisions, component choices, interfaces and assembly constraints.
- Flight firmware timing, sensor integration, PID structure, calibration, saturation and failsafe behaviour.
- Communications, mobile application, target tracking and ROS/Gazebo support, including scope boundaries.

ENGINEERING DECISIONS TO MAKE EXPLICIT

The design review must record data contracts, state transitions, failure handling and why the selected baseline is defensible. Every subsystem should have an interface specification and a reproducible test route.

DESIGN BOUNDARIES AND ASSUMPTIONS

⚠ Recovered code and diagrams do not automatically prove airworthiness or repeatable flight performance.

⚠ Publication rights and sensitive material must be checked before releasing files publicly.

ⓘ Scope is proposed unless explicitly identified as documented existing work. All measured claims require reproducible evidence.

CUSTOM AUTONOMOUS DRONE / PROTOCOL / ACCEPTANCE / REVIEW

EXPERIMENTS, VERIFICATION & PUBLICATION

01

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EXPERIMENTAL METHODOLOGY

01 Reconstruct subsystem requirements and interfaces from original source, schematics, CAD and notes.

02 Recover test evidence: bench measurements, build iterations, flight logs and failure fixes where available.

03 Create software in the loop regression tests for the preserved controller and document any reconstructed assumptions.

MEASUREMENT AND ACCEPTANCE EVIDENCE

- Architecture traceability from source to subsystem diagram
- Clear distinction between verified historical evidence and later reconstruction
- Technical narrative explaining the major engineering tradeoffs

REQUIRED REPOSITORY ARTEFACTS

- Repository README and architecture map
- Curated media pack: photos, PCB renders, CAD and app visuals
- Traceable verification inventory and technical case study

TECHNICAL REVIEW QUESTIONS

Which assumptions could invalidate the result? How do the baselines behave under the same conditions? What is the strongest negative case? Which variables, seeds, workloads, software versions and machine properties must be retained for independent reproduction?

VERIFICATION MAPPING

For each acceptance claim, record the test identifier, input configuration, observed result, pass/fail definition and an independent rerun procedure. Failed or inconclusive cases remain in the report rather than being silently excluded.

PUBLICATION DISCIPLINE

Release original source, configuration files, test harnesses, raw measurements and scripts where rights permit. Distinguish simulation outcomes from physical validation and completed achievements from planned evaluation. Results must not be inferred from targets.